Exact Repeated-Node Cancellation in an Atomic Triangular-Lattice Certificate Uniform Tail Bounds and Jet-Preserving Error Budgets

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Abstract

The assigned manuscript constructs a Gaussian Fourier minorant from finite atomic columns, a summable infinite correction, and whole-parameter sign certificates. Its finite matrices are not the entire certificate. We read the complete cached manuscript and relevant original private bounded-inverse and Bernstein proofs. A source-specific change of columns cancels every periodic repeated-node jet exactly. The counterterms have a closed geometric spectral density, with explicit truncation costs. Keeping the fixed columns and the changed right-hand sides, we prove uniform bounds for the transformed finite and tail systems. Conditional on the source's certified scalar inputs, this yields smaller errors in the original coefficient coordinates and tighter deleted-jet budgets; it does not construct a different global extremizer. The price is that the modified finite columns are no longer purely atomic. Classical Neumann, Fourier, and Bernstein tools are credited, and finite checks, infinite propagation, uniqueness in a chosen coefficient space, and the energy theorem remain distinct. Contribution assessment: C3/5, a source-specific exact repeated-node cancellation and quantified original-coordinate error improvement, not independent certification of triangular-lattice universal optimality.

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1 Scope, normalizations, and current status

Let $b = \sqrt{3}/2$ and

$$A = b^{-1/2}\{m(1,0) + n(1/2, b) : m, n \in \mathbb{Z}\}.$$

This lattice has covolume one. The source uses $\widehat{f}(\xi) = \int_{\mathbb{R}^2} f(x)e^{-2\pi i x\cdot\xi}\,dx$, and A^* is a quarter turn of A. Its energy counts ordered pairs:

$$E_g(\mathcal{X}) = \liminf_{R\to\infty} \frac{1}{N_R} \sum_{\substack{x,y\in\mathcal{X}\cap B_R \\ x\neq y}} g(|x-y|^2), \qquad N_R/(\pi R^2) \longrightarrow 1.$$

Local finiteness and centered disk density are the stated competitor conditions, not uniform separation or translated-disk count bounds. The source claims the triangular minimum for every nonnegative completely monotone g, including infinite energies [1].

Our theorems concern the particular interpolation realization and its norm budgets. The source's all-parameter, all-radius minorant theorem is not a consequence of a finite matrix determinant or of our exact tests. No global proof certification, acceptance determination, or priority claim is made here.

Fresh primary-literature check. Leblé's inspected Theorems 1–2 and Remark 1.1 concern sufficiently small bounded displacements of the hexagonal lattice; the allowed size depends on the interaction, with additional uniform-family conditions [2]. That is not the assigned source's unrestricted competitor claim. Sardari's inspected Theorem 1.7 supplies infinitely many independent radial Schwartz functions with double zeros, together with their Fourier transforms, at the triangular radii [3]. Thus uniqueness in the source's summable coefficient system must not be restated as uniqueness among all radial Schwartz functions. These are inspected primary statements, not complete independent audits of their proofs. The literature refresh used public titles and URLs only; all private corpus material remained local.

2 What the atomic certificate actually verifies

The source has one main scientific TeX file. The following map separates its mechanisms and the evidence used here.

1. Lines 308–650: reciprocal Gaussian duality, a near-zero Fourier mass bound using only centered density, Poisson sharpness, and positive Laplace mixtures. The mixture is of nonnegative energy inequalities, not an asserted Schwartz auxiliary for each mixed potential.

2. Lines 651–1054: modulo-36 nodes, damped periodic columns, rational tail columns, their atomic or absolutely continuous spectral measures, uniform Schwartz realization, exact jets, and direct repetitions.

3. Lines 1057–1400: two order-20 finite blocks, an eight-coordinate containing polytope, omitted target residuals, and finite envelope/sign inequalities. The actual parameter curve lies in the polytope; affine interpolation in the polytope is not interpolation of its exponential parameterization.

4. Lines 1403–1672: all four maps on the declared unweighted ℓ^1 spaces, the own-side repeated-node subtraction, the complete Schur right-hand side, and the infinite correction. Half-sums of the two channel bounds control each original side.

5. Lines 1674–2155: deleted jets on near cells, the exceptional cubic Bernstein argument, a periodic sine-product barrier, and a separate unbounded-range estimate. Ordinary uniform function errors cannot replace the quadratic vanishing at contact nodes.

6. Lines 2158–2310: outward integer interval arithmetic, Gauss–Jordan enclosures, and 659 Bernstein row minima over the whole polytope. These are finite computations, not the preceding analytic arguments.

The source is already explicit about these distinctions. We found no mathematical correction to assert in these mechanisms. In particular, the repeated-node map and the residual term in the Schur equation are present and correctly signed in the source; we do not present their restoration as a new result.

The parent reviewer executed the pinned original finite checker with assertions enabled, reporting exit zero and empty standard output/error. Its receipt is recorded separately. We inspected that receipt, did not duplicate the run, and do not count its 659 source row minima as author tests. The original checker code is outside our literal source-read claim. Our independent exact suite checks the new formulas below.

3 The actual spaces and the private comparison

Put

$$h = 17/50, \quad H = 27/50, \quad \eta = H - h = 1/5, \quad a_0 = \pi\eta, \quad q = e^{-36a_0}, \quad \kappa = \pi/36.$$

The symbol a_0 is a damping exponent, not a lattice vector or a norm bound. Define

$$\begin{aligned} I &= \{0,1,3,4,7,9,12,13,16,19,21,25,27,28,31\},\\ \mathcal{N} &= (36\mathbb{Z} + I) \cap (0,\infty),\\ f &= \{1,3,4,7,9,12,13,16,19,21\}, \quad T = \mathcal{N} \setminus f, \quad L = \{0\} \cup f. \end{aligned}$$

Write $X = \mathbb{R}^{20}$ and $Y = \ell^1(T;\mathbb{R}^2)$, both with sums of absolute values. A pair at a node is ordered (c,d). Every matrix norm below is a maximum absolute column sum or its infinite supremum, never an unstated Euclidean singular value. A uniform column-sum bound gives a bounded map on ℓ^1 by absolute summation and density.

The freshly read original TR147 proposition requires a realized bounded operator on a declared Banach space before invoking Neumann inversion [4]. The original Bernstein chapter proves power conversion, convex-hull enclosure, and exact subdivision, while keeping the analytic remainder separate [5]. These are classical tools already used by the public source, not additional global theorems supplied by the private corpus.

Example 3.1 (Finite inverses do not settle the infinite space)**.** On $\ell^1(\mathbb{N}_0)$, let $(Sx)_j = x_{j+1}$. Every finite leading block of $I - 2S$ is upper triangular with determinant one. Nevertheless $(2^{-j})_{j\geq 0}$ belongs to ℓ^1 and lies in its kernel. Its order-N finite inverse has last-column sum $2^N - 1$. This is the original TR147 warning specialized from ℓ^2 to the present ℓ^1 setting, not a new counterexample credited to this report.

4 Exact cancellation of periodic repeated jets

Set

$$P(s) = \prod_{r\in I}(2\sin\kappa(s-r))^2, \qquad P(n+v) = Q_n v^2(1 + D_n v + O(v^2)).$$

Here $Q_n > 0$ and D_n are 36-periodic. All functions in s denote radial functions under $s = b|x|^2$. A periodic column pair centered at $n \in L$ is

$$F_{n;c,d}(s) = e^{-\pi H s}P(s)\{c\kappa^2\csc^2\kappa(s-n) + d\kappa\cot\kappa(s-n)\}. \tag{1}$$

The removable poles are understood. At a later node $m = n + 36l$, $l \geq 1$, dividing by $e^{-\pi h s}$ gives the jet

$$Q_m e^{-a_0 m}(c, d + (D_m - a_0)c).$$

In rational-tail coefficient coordinates this is $e^{-a_0 m}(c, d - a_0 c)$, with no quotient Q_m/Q_n.

For a list $x = ((c_n, d_n))_{n\in L}$ define $A_L x \in Y$ by

$$(A_L x)_{n+36l} = e^{-a_0(n+36l)}(c_n, d_n - a_0 c_n) \quad (n \in L,\ l \geq 1), \tag{2}$$

and zero on the remaining nodes. The supports for distinct $n \in L$ are disjoint because their residues are distinct.

Proposition 4.1 (Exact alias norm and remainder)**.** *For every real list x,*

$$\|A_L x\|_1 = \frac{q}{1-q}\sum_{n\in L} e^{-a_0 n}(|c_n| + |d_n - a_0 c_n|). \tag{3}$$

Consequently $\|A_L\|_{1\to 1} = (1 + a_0)q/(1 - q)$, since $0 \in L$. If $A_L^{[p]}$ retains only $1 \leq l \leq p$, $p \geq 0$, then

$$\|(A_L - A_L^{[p]})x\|_1 = q^p\|A_L x\|_1.$$

For its restriction $\mathcal{A}$ to f, the exact operator norm is $e^{-a_0}(1 + a_0)q/(1 - q)$.

Proof. Sum absolute values on the disjoint supports and use the geometric series. The induced norm of $(c, d) \mapsto (c, d - a_0 c)$ is $1 + a_0$, attained at $(1, 0)$. The largest $e^{-a_0 n}$ is at $n = 0$ on L and at $n = 1$ on f. Removing the first p terms multiplies every geometric sum by q^p. □

Let the rational-tail synthesis map be

$$\mathcal{T}w(s) = e^{-\pi h s}P(s)\sum_{m\in T}\left(\frac{w_{m,c}}{(s-m)^2} + \frac{w_{m,d}}{s-m}\right).$$

The source's uniform spectral-variation bounds prove its continuity from Y into every radial Schwartz seminorm. This actual realization is essential, not inferred from a formal sum of rational expressions. Define corrected columns by

$$F^{\circ}_{n;c,d} = F_{n;c,d} - \mathcal{T}A_L x_{n;c,d}, \tag{4}$$

where $x_{n;c,d}$ is supported at n.

Theorem 4.2 (Own-side cardinalization)**.** *The function $F^{\circ}_{n;c,d}$ is real radial Schwartz. At every positive interpolation node other than its center its value and slope vanish. At $n \in f$, its jet after division by $e^{-\pi Hs}$ is $Q_n(c, d + D_n c)$. For $n = 0$ it has zero jets at every positive node. Its Fourier transform need not have these zeros.*

Proof. The compensating list is summable by Proposition 4.1, so its synthesis and all derivatives are legitimate. A rational tail column has nonzero direct jets only at its own pole; elsewhere P has a double zero. At $m = n + 36l$ its coefficient pair in (2) gives precisely the divided jet of (1), including $-a_0 c$ in the slope coordinate. Subtraction cancels it. All other positive nodes already have double zeros, and no tail node equals the finite center. This proves all statements, including the retained normalization at n. □

4.1 A closed spectral formula, not a hidden infinite evaluation

Write $P(s) = \sum_{l=-15}^{15} P_l e^{i\pi t_l s}$, $t_l = l/18$. On $t \in (t_{j-1}, t_j)$, $1 \le j \le 15$, define

$$\rho_{c,n}(t) = -2\pi^2 \sum_{l=j}^{15} (t_l - t) P_l e^{i\pi(t_l - t)n},$$
$$\rho_{d,n}(t) = 2i\pi \sum_{l=j}^{15} P_l e^{i\pi(t_l - t)n}. \tag{5}$$

These are the source's folded densities for the undamped rational columns, algebraically meaningful for every zero n of P. Their positive and negative frequencies are paired by conjugation; there is no additional atom at zero.

Proposition 4.3 (Geometric spectral counterterm)**.** *For $n \in L$, the undamped counterterm in $\mathcal{T} A_L x_{n;c,d}$ has folded density*

$$e^{-a_0 n} \frac{q e^{-36\pi i t}}{1 - q e^{-36\pi i t}} \{c\rho_{c,n}(t) + (d - a_0 c)\rho_{d,n}(t)\}. \tag{6}$$

The denominator has modulus at least $1 - q$. Keeping the first p periods replaces the fraction by $\sum_{r=1}^{p} (q e^{-36\pi i t})^r$; its discarded modulus is at most $q^{p+1}/(1 - q)$, uniformly on the full frequency interval.

Proof. For $m = n + 36r$, every summand of (5) acquires the factor $e^{36\pi i(t_l - t)r} = e^{-36\pi i t r}$, because $36 t_l = 2l$. Multiplication by $e^{-a_0 m}$ and summation over $r \ge 1$ give (6). Absolute geometric convergence and the source's uniform variation bounds justify exchanging the finite l sum, the r sum, and the spectral integral. The denominator bound and the finite geometric remainder prove the last assertions. □

Thus each corrected column has an atomic part with damping H and an absolutely continuous part with damping h. They must not be combined under a single unchanged Gaussian damping. The identity is more than a formal coordinate substitution: it exposes how to enclose every compensating period and gives a uniform analytic truncation rule. If the synthesis seminorm satisfies $p_{\beta,\gamma}(\mathcal{T} w) \le C_{\beta,\gamma} \|w\|_1$, then truncating the counterterm after p periods costs at most $C_{\beta,\gamma} q^p \|A_L x\|_1$. This is a proved error bound, not a recommendation to replace the source's all-period equation by an uncertified truncation.

5 The changed finite system and its full right-hand side

Let $\mathcal{A} : X \to Y$ be the restriction above. In the source, $R : X \to X$, $\mathcal{B} : Y \to X$, $\mathcal{C} : X \to Y$, and $\mathcal{D} : Y \to Y$ are the opposite-side maps. Direct own-side repetition is $\mathcal{A}$. For the two sides the exact equations are

$$\begin{aligned} u_1 &= Ru_2 + \mathcal{B}w_2 + r_1, & u_2 &= Ru_1 + \mathcal{B}w_1 + r_2, \\ w_1 &= \mathcal{C}u_2 + \mathcal{D}w_2 - \mathcal{A}u_1 + s_1, & w_2 &= \mathcal{C}u_1 + \mathcal{D}w_1 - \mathcal{A}u_2 + s_2. \end{aligned} \tag{7}$$

The fixed coefficients at zero are $(11/25, 0)$ on both sides, and the fixed direct constants are $C_1 = -3/500$, $C_2 = 3/500$. The s_i already include minus the repeated jets a_i^0 from these fixed zero-centered columns. Set

$$v_i = w_i + \mathcal{A}u_i + a_i^0, \qquad R^\circ = R - \mathcal{B}\mathcal{A}, \qquad \mathcal{C}^\circ = \mathcal{C} - \mathcal{D}\mathcal{A}. \tag{8}$$

For a sign $\epsilon \in \{+1, -1\}$ use subscripts ϵ for the sum of side 1 and ϵ times side 2, including a_ϵ^0. This sign is not the Gaussian parameter or an approximation tolerance.

Theorem 5.1 (Exact de-aliasing with fixed columns)**.** *On $X \oplus Y$, (7) is equivalent to*

$$\begin{aligned} (I - \epsilon R^\circ)u_\epsilon - \epsilon\mathcal{B}v_\epsilon &= r_\epsilon^\circ, & r_\epsilon^\circ &= r_\epsilon - \epsilon\mathcal{B}a_\epsilon^0, \\ (I - \epsilon\mathcal{D})v_\epsilon - \epsilon\mathcal{C}^\circ u_\epsilon &= s_\epsilon^\circ, & s_\epsilon^\circ &= s_\epsilon + (I - \epsilon\mathcal{D})a_\epsilon^0. \end{aligned} \tag{9}$$

Suppose the source bounds, or any rigorously established bounds, give

$$\|\mathcal{B}\| \le B_0, \quad \|\mathcal{D}\| \le d_0, \quad \|\mathcal{A}\| \le a_*, \quad \|\mathcal{C}\| \le c_*, \quad \|(I - \epsilon R)^{-1}\| \le N_\epsilon.$$

If $N_\epsilon B_0 a_* < 1$, define

$$N_\epsilon^\circ = \frac{N_\epsilon}{1 - N_\epsilon B_0 a_*}, \qquad c_*^\circ = c_* + d_0 a_*, \qquad k_\epsilon = d_0 + B_0 c_*^\circ N_\epsilon^\circ. \tag{10}$$

Then $L_\epsilon^\circ = (I - \epsilon R^\circ)^{-1}$ exists with norm at most N_ϵ°. If also $k_\epsilon < 1$, the full system has a unique solution for every $r_\epsilon \in X$, $s_\epsilon \in Y$. Its tail equation is exactly

$$(I - K_\epsilon)v_\epsilon = s_\epsilon^\circ + \epsilon\mathcal{C}^\circ L_\epsilon^\circ r_\epsilon^\circ, \qquad K_\epsilon = \epsilon\mathcal{D} + \mathcal{C}^\circ L_\epsilon^\circ \mathcal{B}. \tag{11}$$

In particular $\|(I - K_\epsilon)^{-1}\| \le (1 - k_\epsilon)^{-1}$ on the full unweighted Y, not merely on finite tail sections.

Proof. Substitute $w_\epsilon = v_\epsilon - \mathcal{A}u_\epsilon - a_\epsilon^0$ in the source channel equations. In the tail equation the own-side term cancels, leaving $-\epsilon(\mathcal{C} - \mathcal{D}\mathcal{A})u_\epsilon$. The constant terms are exactly those in (9); dropping them would not describe the same functions.

The change of variables and its inverse are bounded triangular maps on $X \oplus Y$. Factor $I - \epsilon R^\circ$ as $(I - \epsilon R)[I + (I - \epsilon R)^{-1}\epsilon\mathcal{B}\mathcal{A}]$. The latter factor is invertible by the operator-norm Neumann series, which proves the finite inverse estimate. Also $\|\mathcal{C}^\circ\| \le c_* + d_0 a_*$. Eliminate u_ϵ from (9); the two factors ϵ in the feedback multiply to one, giving (11). Its Neumann series converges in $\mathcal{B}(Y)$ when $k_\epsilon < 1$ and telescopes on both sides. Recovery of u_ϵ, then w_ϵ, and the half-sums is reversible. This proves existence and uniqueness in the specified coefficient spaces only. □

What changes and what does not. The dominant direct repeated-node map is absent from the new tail feedback. Its physical contribution remains in the corrected columns and in the inverse coordinate map. The reconstructed U_i, F_i, J_i are identical to those obtained from the source system by uniqueness. This is a source-specific preconditioning and error improvement, not a new extremal function or an enlarged uniqueness class.

6 A sharper bound in the original coordinates

We now instantiate the construction with source-specific constants. This is conditional on the finite/envelope inequalities in the source, not an independent recomputation of their transcendental inputs. Use the strict source bounds

$$\begin{aligned} B_0 = 340, \quad d_0 = 3 \cdot 10^{-8}, \quad a_* = 2.5 \cdot 10^{-10}, \quad c_* = 7 \cdot 10^{-15}, \\ N_+ = 17, \quad N_- = 5, \quad \rho = 8.2 \cdot 10^{-8}, \quad \sigma = 10^{-26}, \quad k_V = 1.3 \cdot 10^{-7}. \end{aligned} \tag{12}$$

Here a_*, c_* bound the maps on *all* finite pole columns L, including the fixed columns at zero. The restriction to f only improves them. The missing finite target has norm at most ρ, the missing tail target at most σ, and k_V bounds the transformed constant column. These are all bounds in the same coefficient coordinates, not physical function norms.

Let $u_i^0 = (A_i e)_f$ be the source finite approximation, retaining its fixed coefficients. Define $\bar{u}_i$ by solving the *new* finite equations with $v_i = 0$ and the same truncated finite target. This existence statement uses Theorem 5.1; computing a new validated matrix is not needed for the norm estimates. The corresponding function already includes the exactly summed own-side counterterms.

Theorem 6.1 (Uniform original-coordinate improvement)**.** *Assume the source inequalities just specified, including* $\|A_1 e\|_1 < 2.58$, $\|A_2 e\|_1 < 2.80$, *and the valid infinite operator realization. For every actual Gaussian parameter* $\alpha \geq 1$, *the same exact source coefficient solution satisfies*

$$\|u_i - u_i^0\|_1 < 7.84 \cdot 10^{-6}, \qquad \|w_i\|_1 < 1.531 \cdot 10^{-9} \qquad (i = 1, 2). \tag{13}$$

Thus both bounds are improvements over the source's displayed 10^{-5} *and* $3 \cdot 10^{-9}$, *without suppressing the alias counterterm or changing the represented functions.*

Proof. Throughout, channel subscripts mean sums or differences, not maxima of individual-side estimates. The new finite equation applied to $\bar{u}_\epsilon - u_\epsilon^0$ gives

$$(I - \epsilon R^\circ)(\bar{u}_\epsilon - u_\epsilon^0) = -\epsilon \mathcal{B}(\mathcal{A} u_\epsilon^0 + a_\epsilon^0).$$

The total norm of the two original finite pole lists, including zero, is less than six. Hence, with the constants in (10),

$$\|\bar{u}_\epsilon - u_\epsilon^0\|_1 < \Delta_\epsilon := 6 B_0 a_* N_\epsilon^\circ. \tag{14}$$

In particular each changed finite list, including its unchanged fixed entries, still has norm less than three: half the sum of the two Δ bounds below is less than $6 \cdot 10^{-6}$, while the larger original norm is below 2.80.

On these new finite lists, the remaining opposite-side tail map is $\mathcal{C}_L - \mathcal{D} A_L$, whose norm is at most $c_*^\circ = c_* + d_0 a_* = 7.0075 \cdot 10^{-15}$. The direct own-side repetitions have canceled, including those at zero. The transformed constants cancel in the plus channel; their coefficient difference has magnitude .012 in the minus channel. The tail residual bounds are therefore

$$\sigma_+ = \sigma + 6 c_*^\circ, \qquad \sigma_- = \sigma + 6 c_*^\circ + .012 k_V. \tag{15}$$

The finite missing target remains ρ in both channels: it occurs only on side one and is restored, not omitted from the exact system.

Apply (11) to the exact solution minus this modified finite approximation. The corrected right-hand side includes the finite residual, giving

$$V_\epsilon = \frac{\sigma_\epsilon + c_*^\circ N_\epsilon^\circ \rho}{1 - k_\epsilon}, \qquad E_\epsilon = N_\epsilon^\circ (\rho + B_0 V_\epsilon). \tag{16}$$

Then $\|v_\epsilon\|_1 < V_\epsilon$ and $\|u_\epsilon - \bar{u}_\epsilon\|_1 < E_\epsilon$. All quantities in the following table are rational evaluations of (12)–(16), with displayed bounds rounded outward:

quantity	plus channel	minus channel
N_ϵ°	< 17.000025	< 5.000003
k_ϵ	$< 3.004051 \cdot 10^{-8}$	$< 3.001192 \cdot 10^{-8}$
Δ_ϵ	$< 8.670013 \cdot 10^{-6}$	$< 2.550002 \cdot 10^{-6}$
V_ϵ	$< 4.204502 \cdot 10^{-14}$	$< 1.560043 \cdot 10^{-9}$
E_ϵ	$< 1.394246 \cdot 10^{-6}$	$< 3.062073 \cdot 10^{-6}$

Half the sum of $\Delta_+ + \Delta_- + E_+ + E_-$ is less than $7.84 \cdot 10^{-6}$, proving the first bound.

For the original tail, write

$$w_i = v_i - A_L \bar{x}_i - \mathcal{A}(u_i - \bar{u}_i),$$

where $\bar{x}_i$ includes the fixed zero pair and the modified finite pole coefficients. Its norm is below three. Therefore

$$\|w_i\|_1 < 3a_* + \frac{V_+ + V_-}{2} + a_* \frac{E_+ + E_-}{2} < 1.531 \cdot 10^{-9}. \tag{17}$$

The $3a_*$ term explicitly pays for the removed repetitions. It would be incorrect to identify the small v norm with the original w norm. □

The source's coarser plus-channel contraction bound is $3 \cdot 10^{-8} + 340(3 \cdot 10^{-10})17 = 1.764 \cdot 10^{-6}$. The new k_+ bound is smaller, but both are already far below one. The improvement is an explicit certificate budget, not evidence of a comparable runtime speedup or a substantially larger class of lattices. The advantage is modest in physical error even though one block bound changes by several orders of magnitude.

7 Jet-preserving signs and observable residuals

For a C^2 function φ, let

$$U_m[\varphi](v) = \int_0^1 (1-t)\varphi''(m+tv)\,dt.$$

For $v \neq 0$ this is $(\varphi(m+v) - \varphi(m) - v\varphi'(m))/v^2$, and at zero it is $\varphi''(m)/2$. Exact interpolation yields

$$G(m+v) - J_1(m+v) = v^2(U_m[G](v) - U_m[J_1](v)), \qquad J_2(m+v) = v^2 U_m[J_2](v). \tag{18}$$

This source mechanism is retained, not replaced by a plain supremum error. The sign margins vanish quadratically at contact nodes.

Corollary 7.1 (Smaller deleted-jet budgets)**.** *With the same degree-20 source Taylor polynomials and the same imported curvature and Taylor estimates, the exact source functions obey*

$$|U_m[J_i](y\tau) - T_i(\tau)| < \begin{cases} .014, & m = 1,\ y = -1, \\ .0024, & m = 1,\ y = 1, \\ .000122, & m \geq 3, \end{cases} \qquad 0 \leq \tau \leq 1.$$

These replace the displayed source budgets $.018, .003, .00016$.

Proof. The bounds in Theorem 6.1 are measured relative to the *original* source finite interpolant, so its already tested polynomials can be used unchanged. Substitute those bounds into the source curvature rows and retain its Taylor losses:

$$\begin{aligned} 1510(7.84\cdot 10^{-6}) + 26500(1.531\cdot 10^{-9}) + .002 &< .014,\\ 46(7.84\cdot 10^{-6}) + 1160(1.531\cdot 10^{-9}) + .002 &< .0024,\\ 14(7.84\cdot 10^{-6}) + 800(1.531\cdot 10^{-9}) + .00001 &< .000122. \end{aligned}$$

The integral formula for U_m transfers each curvature bound to the whole joining segment, including $v = 0$. No division by an unbounded small-node residual is made. □

7.1 A targeted functional can cost less than the full norm

For a bounded linear observation $\ell_f \delta u + \ell_t \delta w$, where $\ell_f \in X^*$ and $\ell_t \in Y^* = \ell^\infty(T;\mathbb{R}^2)$, the changed coordinates give $p = \ell_f - \ell_t \mathcal{A}$, $q_1 = \ell_t$ and the observation $p\delta u + q_1 \delta v$. Fixed coefficients cancel in a correction. If its residuals in (9) are r, s, then exactly

$$p\delta u + q_1 \delta v = pL_\epsilon^\circ r + (q_1 + pL_\epsilon^\circ \epsilon \mathcal{B})(I - K_\epsilon)^{-1}(s + \epsilon \mathcal{C}^\circ L_\epsilon^\circ r). \tag{19}$$

This follows by substitution, with the same full Schur residual as before. Bounding the two displayed rows in their actual dual norms can retain cancellation lost by multiplying all global norms. For a deleted-jet functional the source's spectral curvature bounds justify boundedness uniformly over the chosen half-cell. A finite evaluation row alone does not establish an ℓ^∞ bound on every tail column. Equation (19) is a useful classical residual identity specialized to these corrected columns; we do not claim a numerical gain without actually enclosing its rows.

8 Finite parameter certificates and uniform limits

For clarity, the source's parameter enclosure is reproduced in its actual coordinates. Put

$$Z = .391, \quad W = .0061, \quad D = .00018, \quad U = .00048, \quad P_* = .0000003.$$

Its data vector is $e = (z, 1, zw, w, zu, u, zp, p)$, with $w = e^{-2/z}$, $u = e^{-3/z}$, $p = e^{-6/z}$ and $z = 1/[\pi(\alpha/b - h)]$. It lies in the containing polytope

$$\begin{aligned} \mathcal{E} = \{(z, 1, Zw - \Delta, w, a, u, r, p) : 0 \le z \le Z, \quad 0 \le w \le W,\\ 0 \le \Delta \le D, \quad 0 \le u \le U, \quad 0 \le a \le Zu, \quad 0 \le r, p \le P_*\}. \end{aligned} \tag{20}$$

The source verifies these inclusions uniformly, including $\Delta = (Z - z)e^{-2/z}$. The simultaneous letters a, u here are polytope coordinates and are unrelated to a tail coefficient pair.

For a row $\lambda = (b_1, c_1, l_1, d_1, k_1, j_1, n_1, r_1)$, its exact minimum over $\mathcal{E}$ is

$$\begin{aligned} m_{\mathcal{E}}(\lambda) = c_1 + Z\min(b_1, 0) + W\min(d_1 + Zl_1, 0) + D\min(-l_1, 0)\\ + U\min(0, j_1, j_1 + Zk_1) + P_*\{\min(n_1, 0) + \min(r_1, 0)\}. \end{aligned} \tag{21}$$

Indeed five free intervals minimize separately, and the (a, u) triangle has vertices $(0, 0), (0, U), (ZU, U)$. This is the source's exact support calculation, not a new optimization theorem.

For polynomial rows t_r, the power-to-Bernstein identity gives $b_i = \sum_{r \le i} \binom{i}{r}\binom{N}{r}^{-1} t_r$. If $m_{\mathcal{E}}(b_i) > \gamma$ for every row, nonnegative Bernstein weights prove $T(\tau; e) > \gamma$ for every $\tau \in [0, 1]$ and $e \in \mathcal{E}$. The

source's 659 row minima are precisely $27 \cdot 21 + 4 \cdot 23$. The exceptional four tests arise by multiplying the rational Gaussian lower comparison by its positive quadratic denominator and taking cubic coefficients in z/Z. The corner $v = 0$ is handled using the actual $z > 0$; it is not a strictly positive constant lower margin on the entire closed (z, v) square.

Proposition 8.1 (Certified row changes)**.** *Suppose a verified row b_i is changed by a known exact row h_i. Define $H_{\mathcal{E}}(h) = \max\{-m_{\mathcal{E}}(h), -m_{\mathcal{E}}(-h)\}$. Then the old lower bound for that row decreases by at most $H_{\mathcal{E}}(h_i)$. With interval row errors, any rigorous upper enclosure of this same expression is sufficient. A remaining analytic remainder must be subtracted separately.*

Proof. The displayed expression is exactly $\sup_{e \in \mathcal{E}} |he|$. Thus $(b_i + h_i)e \geq b_i e - H_{\mathcal{E}}(h_i)$ pointwise. Taking nonnegative Bernstein combinations and then adding a rigorously bounded analytic remainder proves the assertion. □

This classical support-function bound preserves the source's correlated polytope instead of silently replacing it by independently sampled coordinates. It could be used if modified finite columns were recomputed. Our numerical improvement does not require that recomputation: it uses the original sign polynomials through Corollary 7.1.

No hidden passage from finite to uniform. The maps in Theorem 5.1 are independent of z; all parameter dependence is in the data. The constants in Theorem 6.1 are uniform for the actual $\alpha \geq 1$ curve. This provides the necessary operator-norm control as $z \downarrow 0$. The endpoint itself is a limiting coefficient datum, not a finite Gaussian parameter, and the positive normalization $Q_1 z e^{1/z}$ is only used for $z > 0$.

By contrast, on $\ell^1(\mathbb{N}_0)$ the forward shift $S_+ e_j = e_{j+1}$ has $(I - tS_+)^{-1} = \sum_{k \geq 0} t^k S_+^k$ for every $0 \leq t < 1$, but the inverse norm is $1/(1-t)$. Applying it to e_0 proves equality. Pointwise parameter invertibility does not supply a uniform inverse near $t = 1$. The source's strict uniform tail bounds avoid this problem; a family of finite numerical successes would not.

A contact-preserving perturbation is essential. The nonnegative function v^2 becomes negative at its contact point after subtracting any constant $\varepsilon > 0$, however small. More generally, a perturbation with a nonzero value or slope at a contact node cannot be absorbed into a uniform divided-jet error there. If value and slope do vanish and $|h''|/2 \leq \delta$ on the joining segment, then $|h(m + v)| \leq \delta v^2$, exactly the scale in (18). This elementary fact explains why neither the new coordinate system nor a small numerical residual permits skipping the exact infinite interpolation.

9 Costs, retained obligations, and further work

Advantages. The modulo-36 phase identity sums the repeated-jet counterterm without an unbounded list of separate evaluations. The exact norm formula retains the correlation $d - a_0 c$ and gives a geometric truncation certificate. The transformed tail coupling is much smaller, while the proof explicitly accounts for the finite inverse perturbation, the fixed zero columns, and the transformed constants. Most importantly, the final improved errors are stated in the original coordinates, so the source's existing polynomial sign tables remain usable.

Costs and limitations. The source deliberately chooses finite atomic columns. Corrected columns instead have atomic and continuous spectral parts with different dampings. A direct implementation would need validated integration or a further analytic enclosure of (6); no runtime benchmark

or verified replacement finite checker is claimed. The new operator estimates inherit the source's exact finite inequalities and all-node envelope bounds. The conditional improvement does not repair an identified defect: the source already includes the own-side map, full Schur residual, and unbounded-range argument.

The source's Schwartz synthesis relies on uniform variation of its actual tail columns; that hypothesis cannot be replaced by a generic finite Gram matrix. Deleted-jet bounds prove signs only when the interpolation equalities are exact. The separate sine-product barrier is still needed beyond the near cells, and reciprocal Gaussian duality, Poisson sharpness, density-only comparison, and nonnegative Laplace mixtures remain necessary for the energy claim. The original source contains these arguments, but our bounded certificate work is not a claim to have independently certified every global deduction.

Feasible extensions. One can enclose the closed counterterm density on fifteen frequency intervals and compare its actual row norms with the present norm-product bounds. Another option is to cancel only the first p periods, retaining the exact q^p remainder from Proposition 4.1; any resulting sign proof must keep that remainder. These are concrete validation tasks, not completed performance results. A componentwise observable calculation using (19) could exploit additional cancellations, provided every tail column and the full parameter domain are covered.

10 Evidence and assessment

The owned ledger records fresh bounded literal reads of all 2,473 lines of the main TeX, both README files, and the bibliography: 2,743 public text lines in four files. Private original reading comprises TR147 lines 1795–1849 and all 299 lines of the named Bernstein chapter, 354 lines in total. The remaining TR147 material and other private corpus files were not read for this report. Locator hits and parent reading are not added to those intervals. Pre/post hashes and unread complements are recorded independently of parent snapshot authentication.

The exact author suite checks residue coverage, repeated-jet cancellation, geometric density identities, noncommuting block transformations with fixed columns, rational budget comparisons, and finite polytope and Bernstein identities. It passed 77,548 exact assertions in 21,449 bounded cases, with separate units recorded in the owned execution JSON. These bounded checks supplement the proofs; they do not reproduce the original finite checker or certify the source's full Gaussian construction.

Applicability remains L3. The C3 assessment is for the explicit source-specific cardinalization, its closed geometric density, and the proved improvement of the original coefficient and deleted-jet budgets. Classical block elimination, Neumann inversion, support functions and finite exact tests are not individually credited as notable discoveries. There is no C4/C5 claim, new universal-optimality theorem, or verified computational speedup.

References

[1] OpenAI, *An atomic certificate for triangular-lattice universal optimality*, September 26, 2026, assigned cached manuscript, pinned commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Complete four-text literal scope and local hashes are in the owned ledger.

[2] T. Leblé, *The hexagonal lattice is universally locally optimal*, arXiv:2511.03353. Theorems 1–2 and Remark 1.1 inspected in the primary HTML; local bounded-displacement scope, not a complete proof audit.

[3] N. Talebizadeh Sardari, *Higher Fourier interpolation on the plane*, arXiv:2102.08753v2. Theorem 1.7 and its stated reduction inspected; the main construction and all imported modular-form inputs were not independently audited.

[4] M. Subbarao, confidential original Ram-Master manuscript, `RamFoundationTRs/TR-rev-1-old-tr147/TR-rev-1-old-tr147.tex`, lines 1795–1849 freshly read, especially the actual proposition and proof on infinite analytic inversion.

[5] M. Subbarao, confidential original Bernstein and clearance chapter, `RamCalBooks/volumes/volume16/chapters/ch05d-clearance.tex`, all 299 lines freshly read; exact conversion, hull, subdivision and separate analytic remainder obligations.